

TIPS FOR GENETICS PROBLEMS

1. Write down symbols for the alleles. (These may be given in the problem.) When represented by single letters, the dominant allele is uppercase and the recessive is lowercase.
2. Write down the possible genotypes, as determined by the phenotype.
 - a. If the phenotype is that of the dominant trait (for example, purple flowers), then the genotype is either homozygous dominant or heterozygous (PP or Pp in this example).
 - b. If the phenotype is that of the recessive trait, the genotype must be homozygous recessive (for example, pp).
 - c. If the problem says "true-breeding," the genotype is homozygous.
3. Determine what the problem is asking for. If asked to do a cross, write it out in the form [genotype] $\times$ [genotype], using the alleles you've decided on.
4. To figure out the outcome of a cross, set up a Punnett square.
 - a. Put the gametes of one parent at the top and those of the other on the left. To determine the allele(s) in each gamete for a given genotype, set up a systematic way to list all the possibilities. (Remember, each gamete has one allele of each gene.) Note that there are 2^n possible types of gametes, where n is the number of gene loci that are heterozygous. For example, an individual with genotype $AaBbCc$ would produce $2^3 = 8$ types of gametes. Write the genotypes of the gametes in circles above the columns and to the left of the rows.
 - b. Fill in the Punnett square as if each possible sperm were fertilizing each possible egg, making all of the possible offspring. In a cross of $AaBbCc \times AaBbCc$, for example, the Punnett square would have 8 columns and 8 rows, so there are 64 different offspring; you would know the genotype of each and thus the phenotype. Count genotypes and phenotypes to obtain the genotypic and phenotypic ratios. Because the Punnett square is so large, this method is not the most efficient. See tip 5.
5. You can use the rules of probability if a Punnett square would be too big. (For example, see the question at the end of the summary for Concept 14.2 and question 7 below.) You can consider each gene separately (see the section Solving Complex Genetics Problems with the Rules of Probability in Concept 14.2).
6. If the problem gives you the phenotypic ratios of offspring but not the genotypes of the parents in a given cross, the phenotypes can help you deduce the parents' unknown genotypes.
 - a. For example, if $\frac{1}{2}$ of the offspring have the recessive phenotype and $\frac{1}{2}$ the dominant, you know that the cross was between a heterozygote and a homozygous recessive.
 - b. If the ratio is 3:1, the cross was between two heterozygotes.
 - c. If two genes are involved and you see a 9:3:3:1 ratio in the offspring, you know that each parent is heterozygous for both genes. Caution: Don't assume that the reported numbers will exactly equal the predicted ratios. For example, if there are 13 offspring with the dominant trait and 11 with the recessive, assume that the ratio is one dominant to one recessive.
7. For pedigree problems, use the tips in Figure 14.15 and below to determine what kind of trait is involved.
 - a. If parents without the trait have offspring with the trait, the trait must be recessive and the parents both carriers.
 - b. If the trait is seen in every generation, it is most likely dominant (see the next possibility, though).
 - c. If both parents have the trait, then in order for it to be recessive, all offspring must show the trait.
 - d. To determine the likely genotype of a certain individual in a pedigree, first label the genotypes of all the family members you can. Even if some of the genotypes are incomplete, label what you do know. For example, if an individual has the dominant phenotype, the genotype must be AA or Aa ; you can write this as $A-$. Try different possibilities to see which fits the results. Use the rules of probability to calculate the probability of each possible genotype being the correct one.

TEST YOUR UNDERSTANDING

➔ For multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. **DRAW IT** A pea plant heterozygous for inflated pods (Ii) is crossed with a plant homozygous for constricted pods (ii). Draw a Punnett square for this cross to predict genotypic and phenotypic ratios. Assume that pollen comes from the ii plant.
2. A man with type A blood marries a woman with type B blood. Their child has type O blood. What are the genotypes of these three individuals? What genotypes, and in what frequencies, would you expect in future offspring from this marriage?
3. A man has six fingers on each hand and six toes on each foot. His wife and their daughter have the typical number of digits. Remember that extra digits is a dominant trait. What fraction of this couple's children would be expected to have extra digits?
4. **DRAW IT** Two pea plants heterozygous for the characters of pod color and pod shape are crossed. (See Table 14.1.) Draw a Punnett square to determine the phenotypic ratios of the offspring.

Levels 3-4: Applying/Analyzing

5. Flower position, stem length, and seed shape are three characters that Mendel studied. Each is controlled by an independently assorting gene and has dominant and

recessive expression as indicated in Table 14.1. If a plant that is heterozygous for all three characters is allowed to self-fertilize, what proportion of the offspring would you expect to be each of the following? (Note: Use the rules of probability instead of a huge Punnett square.)

- (a) homozygous for the three dominant traits
 - (b) homozygous for the three recessive traits
 - (c) heterozygous for all three characters
 - (d) homozygous for axial and tall, heterozygous for seed shape
6. Hemochromatosis is an inherited disease caused by a recessive allele. If a woman and her husband, who are both carriers, have three children, what is the probability of each of the following?
 - (a) All three children are of normal phenotype.
 - (b) One or more of the three children have the disease.
 - (c) All three children have the disease.
 - (d) At least one child is phenotypically normal.(Note: It will help to remember that the probabilities of all possible outcomes always add up to 1.)
 7. The genotype of F_1 individuals in a tetrahybrid cross is $AaBbCcDd$. Assuming independent assortment of these four genes, what are the probabilities that F_2 offspring will have the following genotypes?
 - (a) $aabbccdd$
 - (b) $AaBbCcDd$
 - (c) $AABBCCDD$
 - (d) $AaBBccDd$
 - (e) $AaBBCCdd$